

emergency SOS button for hospitals when they urgently need blood. it will ping every registered donor in 10 km radius who match the requirements

instead of normal message "The city hospital needs O+ blood" we can give personal urgent message

"Someone 5 km at — needs blood desperate you are a match please help"

← streak and points method

The Donor Dashboard which shows a Streak every three months

give them badges

- first blood donation
- third blood donation
- life saver (who helped SOS patient)
- for every 3 consistent blood donation
- they connect two people they get badge

Problem Statement

A website/app that connects Blood Donor, Recipient and Hospitals to promote blood donation

problem - issue is not lack of blood; its lack of timing and logistics. blood has short shelf time and during emergencies blood finding is chaotic. current it is done through word-of-mouth or unverified social media post

we have three user group

- i) donors
- ii) recipients
- iii) hospital/blood banks

to keep track of blood being used

— when donor donates blood a barcode will be generated and stuck to the bag and when the hospital needs blood the bag is given to them and the barcode is scanned with the hospital registered device and then a message will go to the person your blood saved a life.

15.4 final workflow

Problem Identified

↓
Research

↓
Design UI

↓
Develop Website

↓
Build Database

↓
Develop APIs

↓
Implement AI Search

↓
Barcode Tracking

↓
Emergency SOS

↓
Testing

↓
Soft Deployment

↓
Helping & saving lives ☺

Roll 147 Team Roles

Member	Role	Responsibility
Rounak Prasad	Full Stack developer	Frontend, Backend, Database Integration
Anupam Kumar	Full Stack developer	APIs, Authentication, AI Search, NOTIFICATION
Vivek Lekhak	Presentation lead	PPT Design, UI flow, Documentation
Aditi JHA	Presentation & Design	diagrams, flowcharts, UI Mockups
Rishika Kumari	Research and Review	SIH Research, Feature Validation Testing
Arpita Bhal	Research and Review	Documentation, Literature Survey, Proofreading

13.7 Development Road Map

Requirement Analysis

UI design

Database design

Frontend Development

Backend Development

API Integration

Testing

Deployment

Presentation

12.4 Trust & Verification

Hospital Verification

|

Doctor Verification

|

Donor Verification

|

Blood Bank Verification

|

Verified Badge

|

Trusted Platform

*Website Logic

Hospital uploads Blood Test Result



Website Reads Medical Status



Eligible?



Yes



Keep Active



Available for SOS

No



Mark donor

Temporarily Inactive



Hidden from SOS Searches



Recovery Period Completed



Medical Re-evaluation



Active again

* Digital Blood donation Certificate

After each donation:

Donation Successful



Generate Certificate



Download PDF



Share on Facebook/Instagram



Motivate Others

* Donor Health Verification & Eligibility Flow

Donor visits Hospital



Blood donation Process



Blood Sample Laboratory test



Blood Healthy



Barcode generated



Blood stored



Database updated



Donor Remains
Active

Blood Not Eligible



Test Report generated



Report Sends to donor



Website Receives Status



Donor Status Updated
(Temporarily Inactive)



Doctor / Policy defines
Recovery period



After Recovery → Re-test → Active again

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After Recovery → Re-test → Active again

* Nearby Blood Donation Camps

Hospital Creates Camp



Location Added



User Nearby Notified (VIA whatsapp)



Register



Donate Blood

* Donor Health Tracker

Before donating, donors answer a short health questionnaire

Health form



Weight



Age



Last Donation Date



Medical History



Eligible?



Yes → Donate



No → Suggest Next Eligible date.

Donor donates blood

↓
Donation date saved

↓
90 days completed

↓
Website sends Notification (VIA whatsapp)

↓
"YOUR KINDNESS SAVED lives".
You are now eligible to donate again.
Be someone's hero once more ♥".

* Blood Request Tracking

Patient ~~blood~~ Request Blood

↓
Request Created

↓
Hospital Accepts

↓
Donor assigned

↓
Blood collected

↓
Blood delivered

↓
Request completed

* Live Blood Stock Availability (Hospital Dashboard)

Hospital Updates Blood Stock



Database updated



Website Shows Live Availability



Recipient searches Blood Group



Nearby Hospitals displayed



Available units shown

Example

O D+ : 12 units

O A- : 4 units

O B+ : Out of stock

* Donation Reminder After 3 months

After a donor completes a donation, The system automatically schedules a reminder.

→ consent → health history → pher...

→ In blood testing Phase →

① Compatibility →

→ ABO Grouping } these things remain unchanged, we can save this

→ Rh typing

→ Cross matching

→ Antibody screening

→ Infectious Disease Screening

→ Quality & Component Testing

* For emergency case the doctor sometimes use 'O-' blood, because of the shortage of time.

* Low donor participation also a major problem in hospitals.

We have to balance the supply and demand

Receiver

① If supply ~~increases~~ more than demand, it will be wastage

Receiver have to wait for →

① Finding a donor → depends if the blood is available in hospital

② Transportation of blood → if blood not available in hospital

③ testing and compatibility

④ Quality testing of Donor's blood

of blood increases, cost of Inventory increases

② If demand more than supply, there will going to shortage of blood in hospitals.

What we will solve!

① Reduce ^{time} to find a donor / blood bank

② Manage the supply and demand of blood

③ Increase and make better communication in between donor, hospital and recipient

④ Cut time in different processes

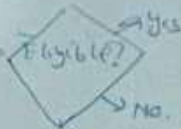
→ consent → health history → physical checks → Eligible?

→ Yes
→ No

We can Hospital and receiver can use their fit like process to find those in emergency.

After

Physical checks →



Hospital Management

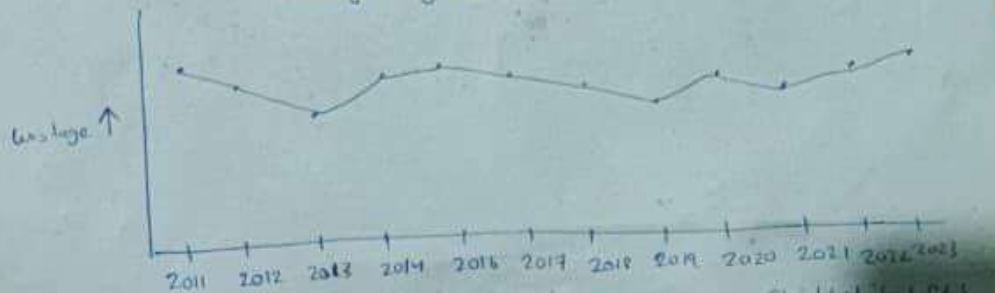
The function of Hospital Management →

- Collect blood
- Compatibility testing
- Inventory control
- Ordering and release
- Data and coordination

We can use AI/ML system in Inventory control of Hospital, to manage bloods and reduce the wastage of blood and increase the availability.

→ In Inventory management, one of the most critical problem is "bullwhip effect" →

In bullwhip effect, when a hospital gets shortage of blood they order more blood from another hospital or blood bank, which creates and increase because of shortage life span of blood it gets expire, and then more the hospital order more blood which creates a hoarding and wastage cycle.



Trends in blood wastage in Taichung Tzu-Chi Hospital, China

Prompt

~~Create a website for blood donation system, in which the donor, hospital and receiver connects can communicate with each other, so that the website works on two different mode (i) Normal mode (ii) Emergency~~
In normal mode - If a receiver wants blood, it

Blood Donation System

It has basically four parts →

- ① Donor recruitment and Donation
- ② Blood-Bank and Hospital management
- ③ Delivery to the right person
- ④

Our main goal is chain this system vein-to-vein

→ Donor Acknow have to do this before donating blood →

- ① Registration
- ② Informed Consent
- ③ Health history
- ④ Physical assessment
- ⑤ Blood collection
- ⑥ Post-donation Counselling

} this process is repeatable, so if we store the donors recent medical records and information, & give to the designated hospital and donor receiver, it will save time and lives.

★ we have to increase the retention of donors.

⑦ We update the data and medical records of donor with recent every time when donor visits hospital.

⑧ Every donor have specific ID by which the other hospital's check the medical history of the donor.